

TRINARY SYMMETRY

Modular Prime Tracks, Eisenstein-Lattice Geometry, and a Bilinear Angular Framework for Prime Distribution

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Preprint status. This manuscript reports a developing research program. It does not claim an unconditional proof of the Riemann Hypothesis. Proved results, computations, geometric models, conditional deductions, and unresolved analytic steps are distinguished explicitly.

Abstract

We develop a framework connecting the two prime-bearing residue classes modulo 6 with Eisenstein-lattice geometry and a bilinear angular Type-II program. The elementary foundation is the fixed-track decomposition $p=6n\pm 1$ for primes $p>3$ and its exact signed multiplication law. A 360-degree wheel and cone/sphere constructions are retained only as geometric motivation. The rigorous formulation uses the Eisenstein integers $\mathbb{Z}[\omega]$, where sixfold unit symmetry, multiplicative norms, ideals, and an exact determinant-angle identity support angular analysis.

The analytic development includes primitive/content decomposition, determinant-strip localization, Smith normal form, regular and singular prime-ideal extraction, recursive large-content scale descent, and a weighted multiplicative-energy estimate for the exact diagonal that is uniform for unequal Type-II lengths. The small-content range is reduced to a two-dimensional additive large-sieve estimate. Its complete bookkeeping remains under audit, so the final Type-II estimate is stated conditionally.

Keywords: prime distribution; congruences; Eisenstein integers; geometry of numbers; large sieve; bilinear forms; Type-II sums.

MSC 2020 (provisional): 11N05, 11N35, 11N36, 11R04.

1. Introduction and scope

Prime distribution is classically studied through congruences, arithmetic functions, zeta and L-functions, sieve methods, and geometry of numbers; see Hardy-Wright [1], Davenport [2], and Iwaniec-Kowalski [3]. This project began from a modular visualization and progressively replaced geometric intuition by exact arithmetic statements.

The 360-degree wheel, cone, double-cone, plane, and sphere constructions are coordinate models, not asserted intrinsic geometries of the integers.

2. Modular prime tracks and wheel

PROVED: Every prime $p > 3$ lies on exactly one fixed track $6n-1$ or $6n+1$.

PROVED: $(6a+s)(6b+t)=6(6ab+at+bs)+st$ for s,t in $\{+1,-1\}$; hence the track sign obeys an exact two-element multiplication law.

MODEL: $x=360q+r$, $0 \leq r < 360$, may be plotted on a 360-spoke residue wheel.

3. Geometric motivation

Successive wheel turns were projected to spiral and cone models, then double-cone, plane, and compactified sphere/inversion pictures. These motivate questions but are not used as proofs.

The absolute track gap is 2 while the relative gap is $1/(3n)$, tending to zero. Compatible residues modulo 6^k instead give a rigorous profinite inverse-system interpretation.

4. Eisenstein formulation

The Eisenstein ring $R=\mathbb{Z}[\omega]$ is a Euclidean domain, PID, and UFD; see Ireland-Rosen [4]. Its norm is $N(a+b\omega)=a^2-ab+b^2$.

PROVED: For $z=a+b\omega$ and $w=c+d\omega$, $[z,w]=ad-bc$ satisfies $|\sin(\arg z - \arg w)| = \sqrt{3} |[z,w]| / (2 \sqrt{N(z)N(w)})$.

PROVED: $Q(q,r)=3q^2+3qr+r^2$ is invariant under $T(q,r)=(q,-3q-r)$. A computational shell label $Q+41$ is not claimed to globally label all primes.

5. Computational observations

COMPUTED: Finite experiments produced symmetric lattice representations, multiple coordinates associated with exponent 2069, recurrence-like six-residue counts, and local multiples-of-6 boundary anchors. These are evidence in tested ranges only.

6. Bilinear angular problem

The Type-II target preserves two variables with $UV=X$.

$$T_M = \sum_{1 \leq |m| \leq M} \left| \sum_a a_u b_v \exp(6im \arg(uv)) W(N(u)N(v)/X) \right|^2.$$

$$\text{Target: } T_M \ll (M+X) X^\epsilon |a|_2^2 |b|_2^2.$$

The target is motivated by large-sieve principles [2,3,5], but is not quoted as a known theorem. A same-sequence fourth-moment route was rejected after angular-concentration counterexamples.

7. Determinant strips and Smith form

With $\alpha=u_1v_1=A+B\omega$ and $u_2=f+g\omega$, $[\alpha,u_2v]=Px+Qy$ where $P=Ag-Bf$ and $Q=A(f-g)+Bg$.

PROVED: For primitive u_2 the coefficient matrix has Smith form $\text{diag}(1, N(u_2))$; hence $d|P,Q$ has lattice index $d^2/\gcd(d, N(u_2))$.

Writing $q=\gcd(P,Q)$ turns the geometric boundary issue into an arithmetic content problem.

8. Small content

AUDIT: Reduced two-dimensional rational frequencies with denominator $d \leq D$ have torus spacing at least D^{-2} . The expected additive large-sieve scale is $V+D^4$, up to logarithmic losses, following the classical spacing philosophy [5].

Taking $D=V^{1/4}$ balances the terms. Complete bookkeeping for primitivity, rational content, Smith multiplicity, smooth norm coupling, and the ramified prime 3 remains the principal analytic audit.

9. Large content and extraction

PROVED: If $\gcd(q, N(u_2))=1$, then qR divides u_1v_1 and the residual scale is X/q^2 .

PROVED: If p^a exactly divides $N(u_2)$, p is not 3, and p^k divides q , then for $p=\pi\pi_{\text{bar}}$ labeled with $\pi^a|u_2$ and π_{bar} not dividing u_2 , $\pi^{(\max(k-a,0))}\pi_{\text{bar}}^k$ divides u_1v_1 .

PROVED: Combining local factors yields $G(q, u_2)|u_1v_1$ with $N(G)=q^2/\gcd(q, N(u_2))$, apart from bounded ramified-prime bookkeeping.

10. Recursive large-content estimate

Allocating G between u_1 and v_1 and summing restricted L_2 norms by divisor incidence preserves a factor $1/N(G)$. Summing $N(G)>Q$ gives a $Q^{-1} X^\epsilon$ contraction. Thus the large-content recurrence closes once the small-content estimate is independently supplied.

11. Exact diagonal

PROVED: For arbitrary coefficients on $N(u)\sim U$ and $N(v)\sim V$, the weighted energy over $u_1v_1=u_2v_2$ is $\ll_\epsilon (UV)^\epsilon \|a\|_2^2 \|b\|_2^2$, uniformly for unequal U, V .

The proof parameterizes solutions as $u_1=gr$, $u_2=gs$, $v_1=sh$, $v_2=rh$ with $(r,s)=1$, then uses Cauchy-Schwarz and Eisenstein divisor bounds.

12. Conditional Type-II conclusion

CONDITIONAL THEOREM: If the fully uniform small-content two-dimensional additive large-sieve estimate is verified with all Eisenstein-specific bookkeeping, then the diagonal estimate and large-content recursion yield $T_M \ll (M+X)X^\epsilon \|a\|_2^2 \|b\|_2^2$ in a genuine Type-II range.

13. Earlier operator framework

An August 2026 manuscript explored a base-3/operator-theoretic route with Sobolev domains, trace-class operators, adelic kernels, and a conditional critical-line architecture. The present paper does not rely on those claims and does not assert an unconditional proof of the Riemann Hypothesis.

14. Cryptographic scope

This work concerns prime distribution, residue structure, lattice geometry, and analytic estimates. It does not present an algorithm for factoring a supplied cryptographic semiprime or decrypting protected information. Future results should nevertheless be screened for unexpected computational implications before public release.

15. AI collaboration and research integrity

The conceptual research program, geometric models, selection of research questions, computational experimentation, and publication decisions are attributed to Jason Frush. Artificial-intelligence systems, including ChatGPT by OpenAI and earlier Gemini-assisted work documented in the prior manuscript, were used as research-support tools.

AI assistance included mathematical dialogue, algebraic expansions and derivation checks, identification of counterexamples to overstrong intermediate conjectures, stress-testing of proof strategies, organization of definitions and appendices, literature discovery, bibliographic formatting, and editorial preparation. AI-generated mathematical steps were treated as proposals requiring checking rather than as authoritative proofs.

AI systems are not listed as authors. Responsibility for mathematical claims, interpretation of computational evidence, verification of references, and dissemination remains with the human author.

16. Open problems

The immediate priority is an independent complete proof of the small-content two-dimensional additive large-sieve estimate. Other open questions include a rigorous bridge, if one exists, between quadratic shell coordinates and the original integer tracks, reproducible large-scale computations, and identifying which geometric constructions encode genuine invariants.

Appendix A. Proof sketches

A.1 Determinant-angle identity

With $\omega = (-1+i\sqrt{3})/2$, direct expansion gives $\text{Im}((a+b\omega)\overline{(c+d\omega)}) = -(\sqrt{3}/2)(ad-bc)$. Since $\text{Im}(z\overline{w}) = |z||w|\sin(\arg z - \arg w)$, the identity follows.

A.2 Smith form

The strip matrix is $\begin{bmatrix} g & -f \\ f & -g \end{bmatrix}$, with determinant $f^2 - fg + g^2 = N(u_2)$. If $\gcd(f, g) = 1$, the gcd of its entries is one, so the Smith invariants are 1 and $N(u_2)$, yielding the index formula.

A.3 Exact multiplicative energy

For $u_1 v_1 = u_2 v_2$, let $g = \gcd(u_1, u_2)$, $u_1 = gr$, $u_2 = gs$, $(r, s) = 1$. Then $v_1 = sh$ and $v_2 = rh$. Two Cauchy-Schwarz inequalities reduce the energy to Eisenstein divisor sums, bounded by X^ϵ on dyadic norm ranges.

A.4 Singular extraction

For primitive u_2 , a nonramified p dividing $N(u_2)$ must split. Localizing the determinant condition forces k powers on the nonsingular conjugate prime and $\max(k-a, 0)$ powers on the singular side. The extracted norm $p^{(2k - \min(k, a))}$ matches the Smith index.

Appendix B. Proof-status ledger

PROVED: modulo-6 tracks and signed multiplication.

PROVED: Eisenstein determinant identity and quadratic involution.

PROVED: Smith index and regular/singular extraction.

PROVED: weighted exact diagonal energy.

DISPROVED for arbitrary coefficients: earlier same-sequence centered fourth-moment target.

AUDIT: complete small-content two-dimensional additive-sieve bookkeeping.

CONDITIONAL: full Type-II angular estimate.

NOT ESTABLISHED: unconditional Riemann-Hypothesis conclusion.

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